Agentic Specification Refinement

Project Proposal | Autonomous Software Engineering | Registered Report | Team Req HQ

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1. Problem and Hypothesis

Product requirement documents (PRDs) are written in plain prose by people who are not thinking about implementation. They usually skip non-functional requirements (NFRs), leave edge cases open, and use words like "fast" or "secure" with no numbers. Turning a PRD into a specification developers can build from is still done by hand, and the gaps tend to surface late as rework. A single LLM prompted to "write the spec" has a different problem: it fills gaps with confident guesses instead of exposing them. We propose a system that finds what is missing and asks about it, rather than inventing it. Our main hypothesis is that an LLM orchestrator working with narrow, fine-tuned small language model (SLM) critics, and a human who is consulted only where it matters, produces more complete specifications than one monolithic model.

2. Approach

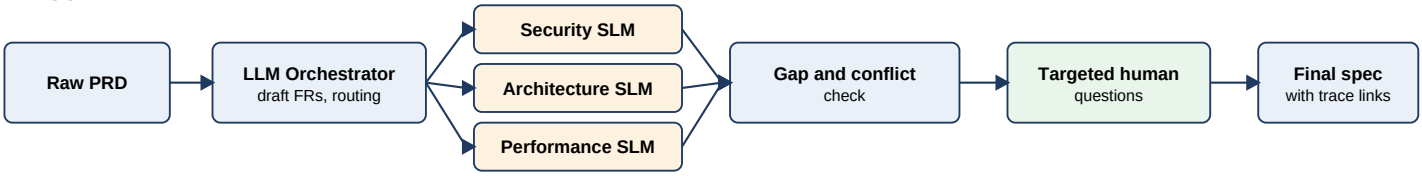

Figure 1: Critics report to the orchestrator. Only unresolved gaps and conflicts reach the human, whose answers go back into the draft spec.

The orchestrator extracts draft functional requirements (FRs) from the PRD. Three SLM critics for Security, Architecture and Performance then review the draft. Each returns missing or vague NFRs and an ambiguity flag that points to the PRD sentence which caused it. The design choice we want to test is that the human does **not** review everything. A gap and conflict check passes a question to the human only when a critic reports missing business context (for example, an expected user load) or when two critics disagree (for example, an encryption requirement that breaks a latency target). Answers return to the orchestrator, and every line in the final spec is traced to a PRD sentence, a critic, or a human answer. We use SLMs because narrow critics can be fine-tuned on labelled NFR data, run locally (PRDs are often confidential), and are cheap enough to run in parallel.

3. Pre-registered Hypotheses

- **H1.** The full pipeline (without human input) has higher NFR recall than a single LLM given the same PRD, with precision dropping by no more than 0.03.
- **H2.** SLM critics come within 0.05 NFR F1 of LLM critics in the same roles, at much lower token cost and latency.
- **H3.** Targeted questioning reaches at least 95% of the completeness of asking the human to review everything, using at least 50% fewer questions.

4. Evaluation Plan

Data. We pair PRD-style proposals with their accepted specifications from open projects (Kubernetes Enhancement Proposals, Rust and Python RFC/PEP histories, and the PURE requirements collection). The PROMISE NFR dataset is used to fine-tune and test the critics. Splits are made by project, and the test set is frozen before any experiment is run. **Ground truth.** Two of us independently extract FRs and NFRs from each reference spec, resolve disagreements, and report Cohen's kappa. **Matching.** Generated and reference requirements are matched semantically using a fixed written rubric, and 20% of matches are checked by hand. **Metrics.** Precision, recall and F1 for FRs and NFRs separately; a completeness penalty, defined as the share of NFR categories (security, performance, scalability, reliability, usability and others) that the PRD text implies but the spec omits; and human questions per PRD. **Comparisons.** (B1) single LLM; (B2) orchestrator with LLM critics; (B3) our pipeline without human input; (B4) our pipeline with ask-everything review; each critic is also removed in turn. Every run is repeated three times, and we report bootstrap confidence intervals with paired tests. The human is simulated by an oracle that answers from the reference spec, to keep trials repeatable.

5. Risks and How We Handle Them

- Reference specs are not perfect truth and may omit things a good spec would include, which understates precision. We report this and check a manual sample of "false" extra requirements.
- Public repositories may be in the training data of the LLMs. We prefer documents dated after the model cutoff and report results on that subset separately.
- A simulated human is more cooperative than a real one, so if time permits we run a small study with classmates answering the questions.
- Negative results will be reported as they are. If H1 fails, the ablations will show which critic or step is responsible.